

Optimizing solar panel tilt angles for enhanced energy Production: Case studies in Lebanon

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ABSTRACT

The solar energy collected and generated by solar panels is influenced by various factors, including geographic location, seasonal variation, system efficiency, and, most notably, the tilt angle of the panels. Among these parameters, the tilt angle is the most controllable, as most solar systems are stationary and operate with an optimal fixed tilt angle. This work presents a numerical method for calculating the optimal tilt angle for monthly, seasonal, and annual use, while also estimating the power generated by solar photovoltaic (PV) systems in different cities across Lebanon. To find the optimal monthly, seasonal, and yearly tilt angles, a MATLAB code was created using mathematical and physical models found in the literature. The code tests and simulates different tilt angle values over the considered period and identifies the angle that maximizes the power generated by the solar panels. The results obtained are compared with those available in the literature and findings show that using monthly or seasonal optimal tilt angles increases the power generated by solar systems by approximately 4–6 %. Additionally, two case studies for a domestic house and a water pump station were present to better illustrate the tilt angle effect on cost savings and carbon emissions reduction compared to the electricity from national grid and carbon emissions from fossil fuel. The results demonstrate a reduction in system costs when implementing a monthly variable tilt angle, along with a decrease in carbon emissions.

1. introduction

The necessity for a variety of energy generation and conversion approaches has been highlighted by the rise in power consumption brought on by population growth and industrial development. Academics are concentrating more on enhancing energy systems and optimizing energy management techniques due to the growing energy needs needed to power everyday activities, which range from sophisticated medical equipment and data processing to transportation and telecommunication [1–3]. The issues of striking a balance

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between rising consumption and sustainability, dependability, and efficiency are the focus of this increased emphasis [4,5]. Researchers are striving for solutions that satisfy the many and changing demands of contemporary society while guaranteeing efficient and responsible energy use by developing technology and improving energy distribution and consumption practices [6–8].

Among renewable energy resources, solar energy stands out as the most prevalent and widely adopted [9]. Between 2021 and 2022, the global solar energy installed capacity have shown an increase of 22 % approximately, increasing from 861 MW in 2021 to 1053 MW in 2022 [10]. Photovoltaic (PV) solar panels were shown to be among the most utilized solar energy technologies worldwide. Solar insolation is harvested by the upper surface of a PV panel to generate electricity. However, its efficacy could be hindered by different environmental factors. Fig. 1 shows the different environmental factors affecting the PV electric conversion efficiency.

To address these challenges, innovative technologies like solar tracking systems, panel cleaning mechanisms, and cooling systems have emerged, aiming to bolster the efficiency of solar PV systems [12,13].

The literature of the utilization of solar panels have shown two modes of installment, fixed and tracking PV systems [14]. Fixed solar panel systems are frequently utilized since they have lower installation and maintenance costs than other systems. When constructing a solar panels system, the tilt angle is considered one of the most critical characteristics to take into account for the system to ensure maximum solar radiation harvesting. The tilt angle is characterized by being the angle between the solar PV panel and the horizontal or vertical plane. Its optimum value depends on the geographical latitude, sun's position, and the climatic conditions mainly, along with other factors [15]. Most systems are designed and implemented for year-round use, therefore determining the tilt angle must take into account these parameters as well as the constraints present throughout the year.

For instance, an off-grid system for residential use should optimize power generation in the winter, whereas an on-grid system must optimize power generation throughout the year. This is different for solar power systems that are used seasonally, like water pumping systems that are only used in the summer. Therefore, the range of days the system will be operated is one of the other crucial elements for determining the ideal tilt angle. The ideal tilt angle for daily use at home varies from the ideal tilt angle for each month, which varies from the ideal tilt angle for a certain range of days or months.

Different studies were conducted in the literature on determining the optimum tilt angle of a solar panel. Liu et al. [16] provided a framework to find the optimal tilt angle of PV systems incorporating satellite with ground data. Results have shown a high correlation between the optimal tilt angle and the diffuse fraction and latitude with a correlation coefficient R of 0.8 and 0.86 respectively. Additionally, an average of 13.7 % increase in the annual power yield was noticed when the tilt angle of the PV panels if fixed at an optimum value compared to horizontally fixed PV panels. Moreover, Prunier et al. [17] numerically investigated the PV tilt angle optimization for maximized energy production for both short-term and permanent PV installations. Results have shown an increase in the annual energy production of 3 %–4.8 % with 2 reorientations per year for all studied locations compared with no reorientations. Additionally, Rinchi et al. [18] performed an analysis on predicting optimal tilt angles for PV systems using comprehensive data-driven approach via machine learning. Results have shown that dividing meteorological features to monthly resolution with the utilization of optimized Multilayer Perceptron model, a root mean squared error down to 1.029° is reached with an accuracy reaching 99.27 %. Abdelaal et al. [19] numerically investigated the PV panels optimal tilt angle. Results have shown that the maximum solar radiation is produced with daily optimal tilt angle alteration, however, this was shown to be impractical and costly. The use of two optimal tilt angles per year for Suez city, results with the best results, which have very near values to optimal tilt angle daily alteration. Consequently, Alam et al. [20] used MacCormach explicit finite difference technique to forecast solar panels optimum tilt angle in India. Results have shown that a 1 % loss of energy was present with seasonal tilt angle variation, compared to a 7 % loss with annually fixed tilt angle. Furthermore, Gürfidan et al. [21] numerically investigated the solar panels optimal tilt angle for improved electric efficiency and energy cost reduction in Turkey. Accuracy was shown to reach 99.0549 % by the developed ExtraPSO model. Results have shown a 29° optimum angle for Turkey throughout the year. Alongside this, Xu et al. [22] proposed numerically a model for soiled PV panel optimum tilt angle. Results have shown that the tilt angle increases to 25.89° – 26.06° when considering dust effect. In addition, Arslan et al. [23] experimentally studied PV panel's azimuth and tilt angles for maximized solar radiation gain. Results have shown an optimal tilt angle of 32.08° with the horizontal plane, with a 0° azimuth angle for Konya province in Turkey. Ashetehe et al. [24] developed generic models to predict the annual and seasonal PV tilt angle without the use of meteorological data at any location in Ethiopia. Results have shown that for seasonal optimal tilt angle, a 5.723–7.532 % gain in total solar radiation was noticed for isotropic model and 9.025–25.796 % total solar radiation gain using anisotropic models. Moreover, Mamun et al. [25] performed indoor and outdoor experiments to investigate the effect of PV tilt angle on their performance and electric parameters under Malaysian climatic conditions. Results have shown 2.09 and 3.45 W drop in output power for indoor and outdoor setups respectively, when increasing the tilt angle by

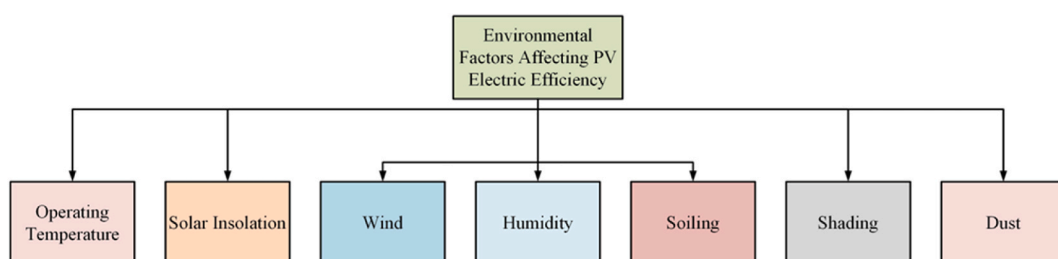


Fig. 1. Environmental factors affecting PV electric output [11].

5°. Moreover, 0.54 % drop in efficiency was noticed for indoor setup when increasing the tilt angle from 0° to 15°, while the drop in efficiency was 0.76 % at outdoor setup for the same increase in tilt angle.

Talebizadeh et al. [26] determined the daily, monthly, seasonal, and yearly optimum slope angles of solar collectors for various regions of Iran. Additionally, new correlations were developed to calculate the monthly, seasonal, and yearly optimum slope angles for latitudes ranging from 20° to 40° North. Additionally, Khosravi et al. [27] presented new correlations for determining the monthly optimum slope angle (OSA) for flat-plate collectors. Twelve equations were developed using a linear regression model to calculate the monthly OSA for stations in the Northern Hemisphere (from 15° to 55°) and the Southern Hemisphere (from −20° to −45°). Additionally, a new equation for calculating the yearly tilt angle was developed and compared with several other methods from the literature. The results confirmed a 20 % increase in solar energy absorption by adjusting the collectors' tilt angle on a monthly basis. Sassine et al. [28] studied the optimal positioning for solar sensors in Beirut across three configurations: a constant optimal angle, a monthly variable tilt angle, and a solar tracking system. The results indicate that the optimal angle of inclination for Beirut is approximately 30° (27.9°). By varying this angle monthly using an adjustable tilt mechanism, a 6 % increase in collected energy was achieved compared to using a constant optimal angle. Moreover, a set of 12 equations were provided by Nijegorodov et al. [29] to calculate the monthly optimum slope angles for any location with latitudes ranging from 60° south to 60° north. Furthermore, Lunde [30] and Garg et al. [31] used $\beta_{opt} = \phi \pm 10$ to estimate the solar collector optimum tilt angle.

As seen in the literature of finding the optimal angle for PV panels, the studies were conducted on different cities around the world with few studies addressing the Middle East and specifically Lebanon. The present work aims on studying the optimal PV tilt angle for three different coastal cities in Lebanon: Beirut, Tyre, and Tripoli. Taking into account the geographical map of Lebanon, Beirut is located at the central point of the coastal line while Tyre lies to the south of the coastal line and Tripoli is situated to the north. Monthly, seasonal, and yearly optimum tilt angles were studied and compared with existing literature. Additionally, the current study presents a case study, featuring an economic analysis and CO₂ emission reduction assessment for a domestic house and a water pump station in the aforementioned three cities.

The originality of the current work lies in following key aspects.

- **Accurate and efficient computation:** The developed tools provide a significant improvement through offering fast numerical computations with high accuracy, securing reliable results for PV tilt angle application.
- **Global applicability:** The developed tools could be utilized at any geographical location through inputting the geographical coordinates of the intended location, making them highly versatile and flexible.
- **Dynamic tilt angle optimization:** The tools can calculate the optimal tilt angle for any period of the year, depending on the specific application of the solar panels system.
- **Addressing tilt angle research gap in the region:** This work provides numerical values and case studies for cities in Lebanon, a region with limited literature coverage on city-specific data, tilt angles, and time periods. These cities specific numerical case studies contribute to a better understanding of tilt angle optimization in a region where such data is scarce.
- **Practical implementation:** The results of this study offer insights on solar energy efficiency through practical case studies.

By addressing these key aspects, this study establishes a strong foundation for further research in solar energy optimization.

2. materials and methods

2.1. Solar angles

The sun's angular locations in the sky with respect to a particular spot on the Earth's surface are known as solar angles. Because they directly affect the angle at which the sun's rays reach the Earth, these angles are important in determining the amount of solar

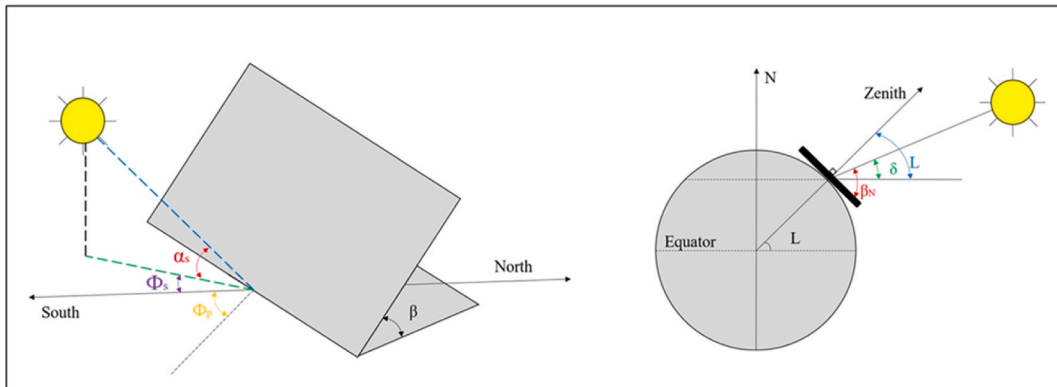


Fig. 2. Solar angles for a solar panel [32].

irradiance received at that location. Solar angles are crucial to the design of solar power systems because they determine how much solar insolation a specific area receives at any one moment. By accurately measuring these sun angles, engineers can strategically optimize the orientation and tilt of solar panels to maximize the electricity generated from the available solar radiation. This optimization is critical for improving the overall efficiency and performance of solar power systems. Fig. 2 represents the different solar angles for a solar panel.

a Inclination angle (α_s)

The angle formed by the sun's light and its projection on a horizontal surface is known as the inclination angle. Determining a sun ray's position in relation to a horizontal surface requires an understanding of the inclination angle, which has uses in solar energy systems and architectural design. The solar altitude angle can be computed using [33]:

$$\sin \alpha_s = (\sin L \times \sin \delta) + (\cos L \times \cos \delta \times \cos h) \quad (1)$$

Where α_s is the inclination angle, L is the angle of latitude, δ is the declination angle, and h is the hour angle.

b Zenith angle (θ_z)

The angle formed by the sun's beams and the horizontal plane's normal is known as the zenith angle. Prior to solar noon, it is positive; during solar noon, it approaches zero; and following solar noon, it becomes negative. Understanding the sun's position in relation to a particular spot on the Earth's surface depends on the zenith angle's movement during the day, which affects solar energy generation. The zenith angle can be calculated using [33]:

$$\theta_z = 90^\circ - \alpha_s \quad (2)$$

Where θ_z is the zenith angle.

c Solar azimuth angle (ϕ_s)

The sign of the solar azimuth angle, which is a horizontal plane angle that is measured from the south to the sun's ray projection, varies depending on the direction; it is negative from the south to the east and positive from the west. It can be computed using [33]:

$$\sin \phi_s = \frac{(\cos \delta \times \sin h)}{\cos \alpha_s} \quad (3)$$

Where ϕ_s is the solar azimuth angle.

d Angle of incidence (θ)

The angle of incidence represents the angle between the sun's rays and the flat surface's normal. It is equivalent to the zenith angle when the surface is horizontal. [34]. This angle is vital in determining how direct sunlight falls on a surface. The angle of incidence can be calculated using [34]:

$$\cos \theta = \sin L \sin \delta \cos \beta - \cos L \sin \delta \sin \beta \cos \phi_p + \cos L \cos \delta \cos h \cos \beta + \sin L \cos \delta \cos h \sin \beta \cos \phi_p + \cos \delta \sin h \sin \beta \sin \phi_p \quad (4)$$

Where ϕ_p is the surface azimuth angle, β is the tilt angle, ϕ_p is the surface azimuth angle.

e Surface azimuth angle (ϕ_p)

The surface azimuth angle, measured on the horizontal plane, is formed between the line pointing south and the horizontal projection of the normal to the inclined surface [35]. When measured from the south towards the west, it is considered positive. This angle is essential for understanding the orientation of an inclined surface relative to the south and plays a crucial role in solar energy systems and architectural design.

f Tilt angle (β)

The tilt angle of a solar panel denotes the angle between the panel's surface and the horizontal plane, representing the panel's slope [20]. This angle is vital in optimizing the panel's orientation to maximize the solar energy capture for solar power systems.

g Sunset/sunrise hour angle (h)

The angle at which the sun moves across the sky represents the sun hour angle. It is measured from the solar noon position. It could

be computed using [36]:

$$H = (t - 12) \times 15^\circ \quad (5)$$

Where H is the solar hour angle and t is the local solar time.

h Declination angle (δ)

The declination angle is the angle between the sun's position and the earth's equator. This angle is critical for understanding the sun's position in the sky and plays a significant role in various fields such as astronomy and solar energy applications [37].

$$\delta = 23.45^\circ \sin \left\{ \frac{360}{365} (284 + N) \right\} \quad (6)$$

Where N is day number were, January 1 being the day number 1.

2.2. solar irradiance

The amount of solar energy collected by the PV module has a direct impact on the power produced by photovoltaic (PV) systems. Nonetheless, the PV module can be positioned with varying tilt angles and orientations. Estimating the solar irradiance on tilted surfaces can be done by combining the components of beam, diffuse, and reflected solar irradiance on the tilt surface. This approach allows for an accurate calculation of solar energy on tilted PV modules, even when only horizontal surface data is available. The total solar irradiance on tilt surface is given by Refs. [37,38]:

$$I_t = I_{bt} + I_{dt} + I_{rt} \quad (7)$$

Where I_t is the total solar irradiance on the tilt surface, W/m^2 ; I_{bt} is the total beam solar irradiance absorbed by the tilt surface, W/m^2 ; I_{dt} is the total diffuse solar irradiance absorbed by the tilt surface, W/m^2 ; and I_{rt} is the total reflected solar irradiance absorbed by the tilt surface, W/m^2 .

a Beam solar irradiance

The direct-beam irradiance, denoted as I_b (perpendicular to the rays), can be translated into the beam irradiance that strikes a PV module's face, represented as I_{bt} . This transformation can be achieved through a straightforward function of the angle of incidence (θ). The angle of incidence is the angle formed between a line drawn perpendicular to the PV module's face and the incoming beam irradiance. The relationship between these quantities is expressed by a mathematical formula [39]:

$$I_{bt} = I_b \times \cos \theta \quad (8)$$

where I_b is the beam portion of the radiation reaching the earth's surface (normal to the rays), and is given by Ref. [39]:

$$I_b = A \times e^{-Km} \quad (9)$$

$$A = 1160 + 75 \times \sin \left\{ \frac{360}{365} (N - 275) \right\} \quad (10)$$

$$K = 0.174 + 0.035 \times \sin \left\{ \frac{360}{365} (N - 275) \right\} \quad (11)$$

$$m = \frac{1}{\sin \alpha_s} \quad (12)$$

Where A is the apparent extraterrestrial flux (W/m^2), K is a dimensionless factor called optical depth, m is air mass ratio, N is day number where January 1 being day 1, and α_s is the inclination angle.

b Diffuse solar irradiance

Assuming an isotropic sky, the fundamental models of diffuse irradiation assume that it arrives at a site with uniform intensity from all directions. In fact, however, on cloudy or hazy days, the sky is brighter close to the sun, and measurements show a comparable effect on clear days. These intricacies are frequently disregarded in spite of this.

The provided expression represents the calculation for diffuse irradiance on a photovoltaic module [39].

$$I_{dt} = C \times I_b \times \left(\frac{1 + \cos \beta}{2} \right) \quad (13)$$

$$C = 0.095 + 0.04 \sin \left\{ \frac{360}{365} (N - 100) \right\} \quad (14)$$

Where C is the sky diffuse factor.

c Reflected solar irradiance

Radiation reflected by surfaces in front of a photovoltaic module is the final source of irradiance that reaches the module. The simplest model takes into account a large horizontal region in front of the module, with a diffuse reflectance (ρ) that uniformly distributes the reflected irradiance in all directions. The formula provided represents the calculation for solar reflected irradiance on the photovoltaic module [40].

$$I_{rt} = \rho \times I_b \times (C + \sin \alpha_s) \times \left(\frac{1 - \cos \beta}{2} \right) \quad (15)$$

Where ρ is the reflectance of the horizontal area.

2.3. simulation

The power produced per square meter of PV panel area (monthly, seasonally, and annually) at beta angles ranging from 0 to 70° was determined using a MATLAB simulation in the current investigation. The ideal tilt angles for the three cities of Tyre, Tripoli, and Beirut were determined by the highest power output the PV systems produces. This optimum angle is noted for every month, season, and for the entire year. The power generated per square meter in these three cities for the monthly, seasonal, and yearly optimum tilt angles are presented further in part 3 of the manuscript.

The MATLAB code requires the city's geographic location, slope angle (β), and time as inputs. The code uses these inputs then to determine various sun angles, with the angle of incidence θ being the most critical angle. These angles are then used to compute the generated power of the solar system, which depends on the value of the tilt angle β , with the efficiency of the solar panel system assumed to be constant at 20 % [41]. The maximum output power for β_{opt} is determined by varying β from 0° to 70° and recording the output power at each value. Fig. 3 denotes the flow diagram of the MATLAB code executed.

3. results and analysis

The calculated monthly, seasonal, and yearly optimum angles are compared with values obtained from other equations in the literature.

Fig. 4 represents a comparison of the monthly optimum tilt angles for the three cities, calculated using the current MATLAB code (β_{opt}) and those calculated using the Nijegorodov et al. [29] equations (β_N). As seen in Fig. 4 (a) in Beirut, the tilt angle calculation showed average values of 31.25° and 32.67° for both β_{opt} and β_N respectively. While in Tyre (b), the tilt angle calculation showed average values of 30.91° and 32.16° for both β_{opt} and β_N respectively. Additionally, the calculation of the tilt angle in Tripoli (c) showed average values of 31.58° and 33.15° for both β_{opt} and β_N respectively. This contributed to an average difference between the β_{opt} and β_N of 4.54 %, 4.04 %, and 4.97 % for Beirut, Tyre, and Tripoli respectively.

Fig. 5 represents a comparison between the seasonal optimum tilt angles calculated using the current MATLAB code (β_{opt}) and those calculated using the Garg (β_G) and Lunde (β_L) equations [30,31], for Beirut, Tyre and Tripoli. As shown in Fig. 5, β_{opt} , β_G , and β_L showed the same trend for the three cities studied, having maximum values during Winter and Fall and minimum values during Summer and Spring, with minimal differences when comparing the tilt angle calculated using a proposed method between the cities. In Beirut, Tyre, and Tripoli, the optimum tilt angle (β_{opt}) showed average values of 30.75°, 30.25°, and 31.25° respectively. While tilt angle calculated

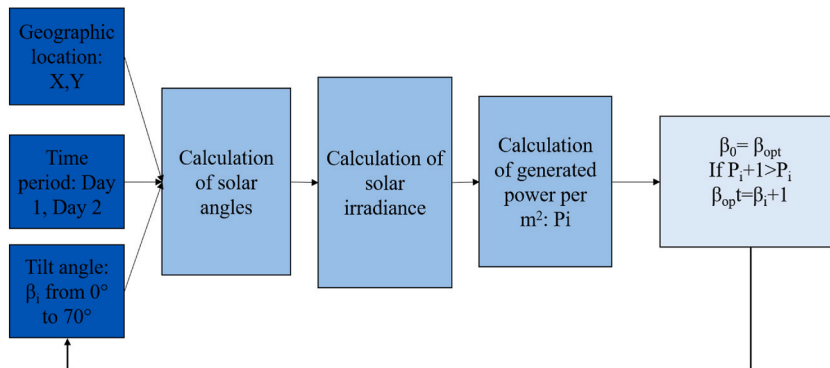


Fig. 3. MATLAB code flow diagram.

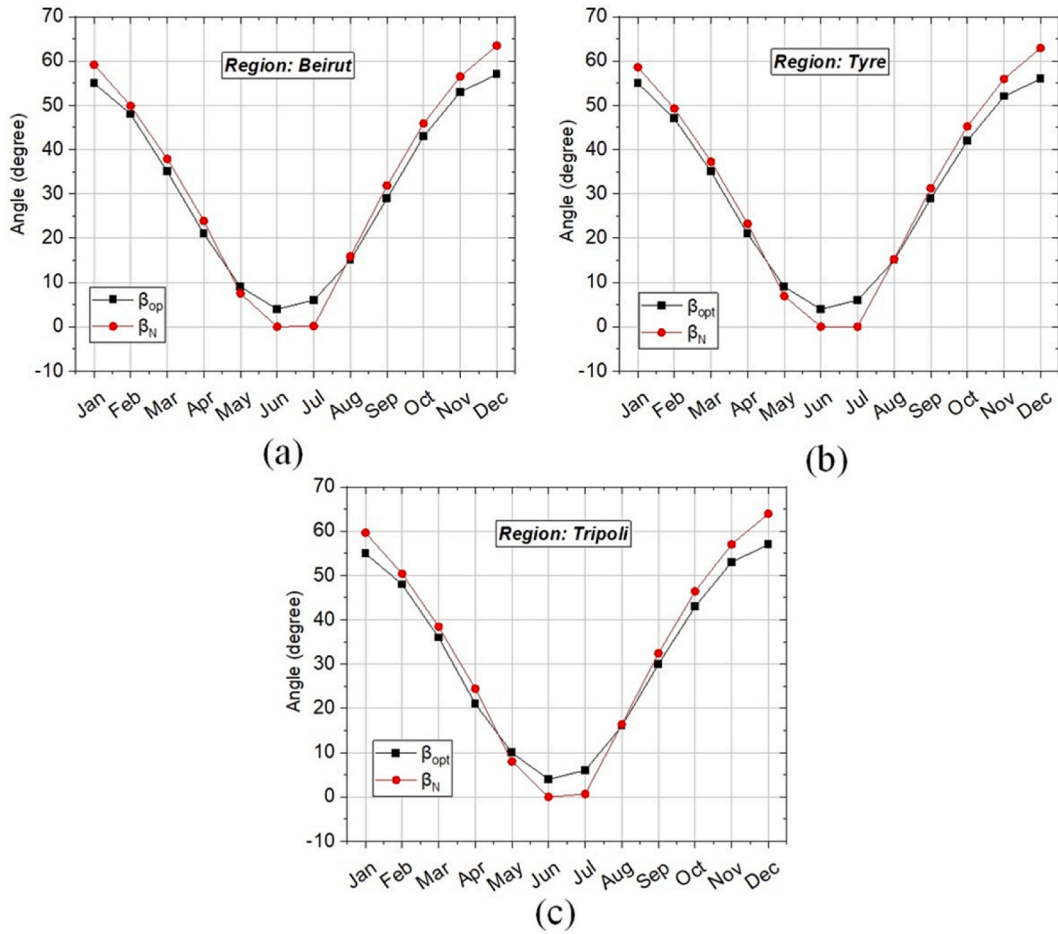


Fig. 4. Comparison of monthly optimum tilt angles for (a) Beirut, (b) Tyre, and (c) Tripoli.

using Garg (β_G) and Lunde (β_L) showed values of 33.89° , 33.26° , and 34.44° in Beirut, Tyre, and Tripoli respectively, showing a difference of about 3° ($\sim 10\%$) between the values calculated using the proposed method (β_{opt}) and the Garg (β_G) and Lunde (β_L) methods.

Table 1 provides a comparison between the yearly optimum tilt angles calculated using MATLAB code and those calculated using the Heywood [42] equation.

As shown in Table 1, the tilt angle difference between the proposed calculation (β_{opt}) and that of Heywood ($\beta_{opt \text{ Heywood}}$) was approximately up to 17 %.

The difference between the calculated angle present in this work and those from the literature was due to the fact that the models from the literature were average models. The difference was shown to be significant for the yearly and seasonal angles, whereas it was smaller for the monthly optimum angle, where the models were monthly adjusted.

Fig. 6 compares the energy generated per square meter for the monthly variable optimum tilt angle to the amount achieved with a constant optimum angle across the year, computed using the MATLAB code. As shown in Fig. 6 (a), a total difference of around 31.37 kWh/m^2 per year between the monthly variable and fixed optimum tilt angles was shown for Beirut. While in Tyre (Fig. 6 (b)), a total difference of around 29.54 kWh/m^2 per year between the monthly variable and fixed optimum tilt angles was shown. In Tripoli (Fig. 6 (c)), a total difference of around 29.34 kWh/m^2 per year between the monthly variable and fixed optimum tilt angles was shown.

Similarly, Fig. 7 shows the power generated per square meter for the seasonal optimal tilt angle, and compares it to the generated value for a constant yearly optimum angle. A total gain of around 24.22, 24.33, and 25.13 kWh/m^2 was shown per year for the seasonal optimum angle was shown for the case of Beirut, Tyre and Tripoli, respectively, compared to a fixed optimum angle over the year.

4. case studies

4.1. solar panel system for home use

To illustrate the significance of employing variable seasonal and monthly optimum tilt angles, consider a home situated in each of

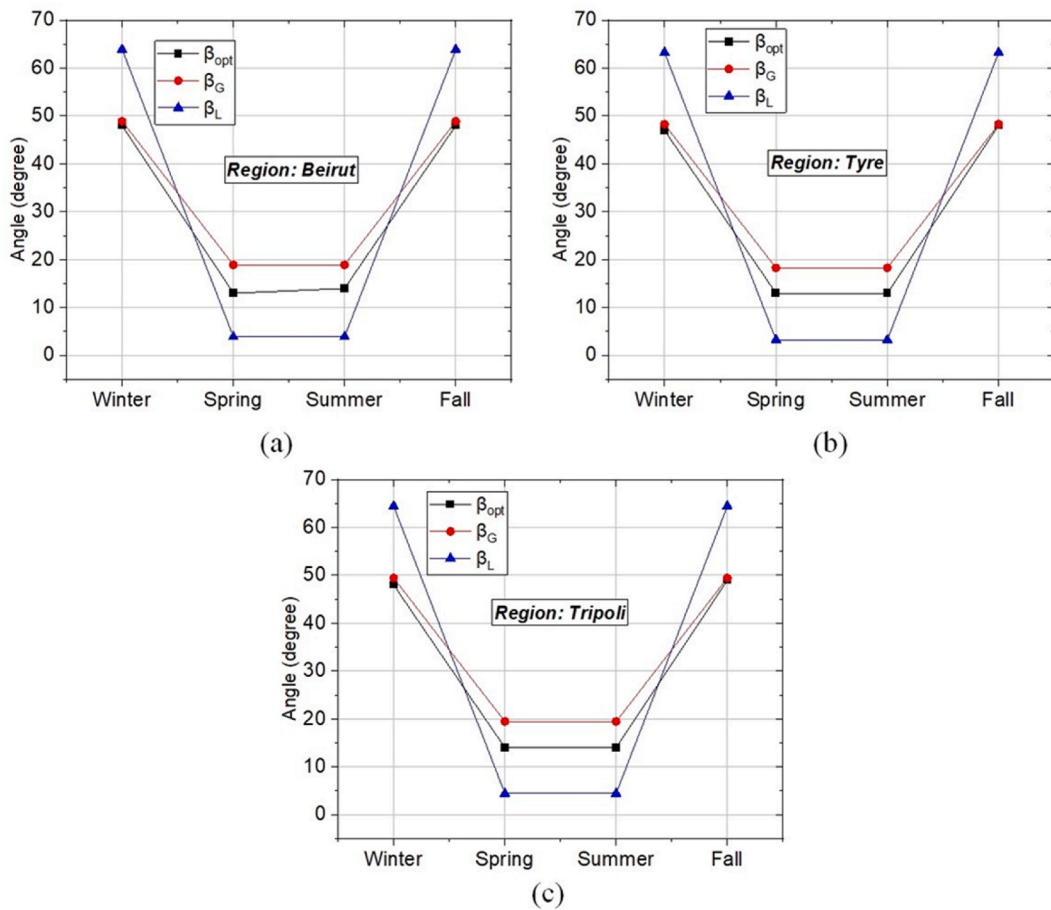


Fig. 5. Comparison of seasonal optimum tilt angles for (a) Beirut, (b) Tyre, and (c) Tripoli.

Table 1

Yearly optimum tilt angles of the three studied cities.

		Beirut		Tyre		Tripoli	
		$\beta_{opt,y}$	β_{opt} Heywood	$\beta_{opt,y}$	β_{opt} Heywood	$\beta_{opt,y}$	β_{opt} Heywood
year mean	365	28	23.89	27	23.26	28	24.44

the three cities Beirut, Tyre, and Tripoli, with an estimated annual electricity consumption of 5000 kWh. Assume that CO₂ emissions from fossil fuel generation amount to 0.5 kg per kWh and that CO₂ emissions for 1 m² of solar panels are 10 kg per year [43], with a panel lifespan of 20 years. Furthermore, consider that one kWh costs \$0.20 from the national electrical grid. Table 2 represents the cost required to generate 1 kWh using variable monthly, seasonal, and fixed yearly optimum tilt angles compared to the cost of 1 kWh from the national electricity grid.

The results indicate that using a variable optimal tilt angle lowers the cost of a 1 kWh by around 90 % when compared to national grid electricity, and by about 0.5 % when comparing a monthly variable angle to an annual fixed angle. In terms of emissions, using a solar power system reduces emissions by 96 % compared to electricity from the national grid, and by 0.36 % when comparing a variable tilt angle to a fixed tilt angle. If a 0.6 % annual deterioration in panel efficiency is considered [44], the cost of 1 kWh is reduced to approximately 84 % compared to national grid electricity, and emissions are reduced by around 90 %. Furthermore, if dust accumulation on the solar panels is taken into account—resulting in an estimated 10 % reduction in efficiency [43]—the cost of 1 kWh drops to about 76 % of the national grid cost, and emissions are reduced by approximately 81 %.

4.2. Case 2- water pump station

Consider the case of a water pump station for a constant load that works during the summer from 21 Jun to 23 September, and the

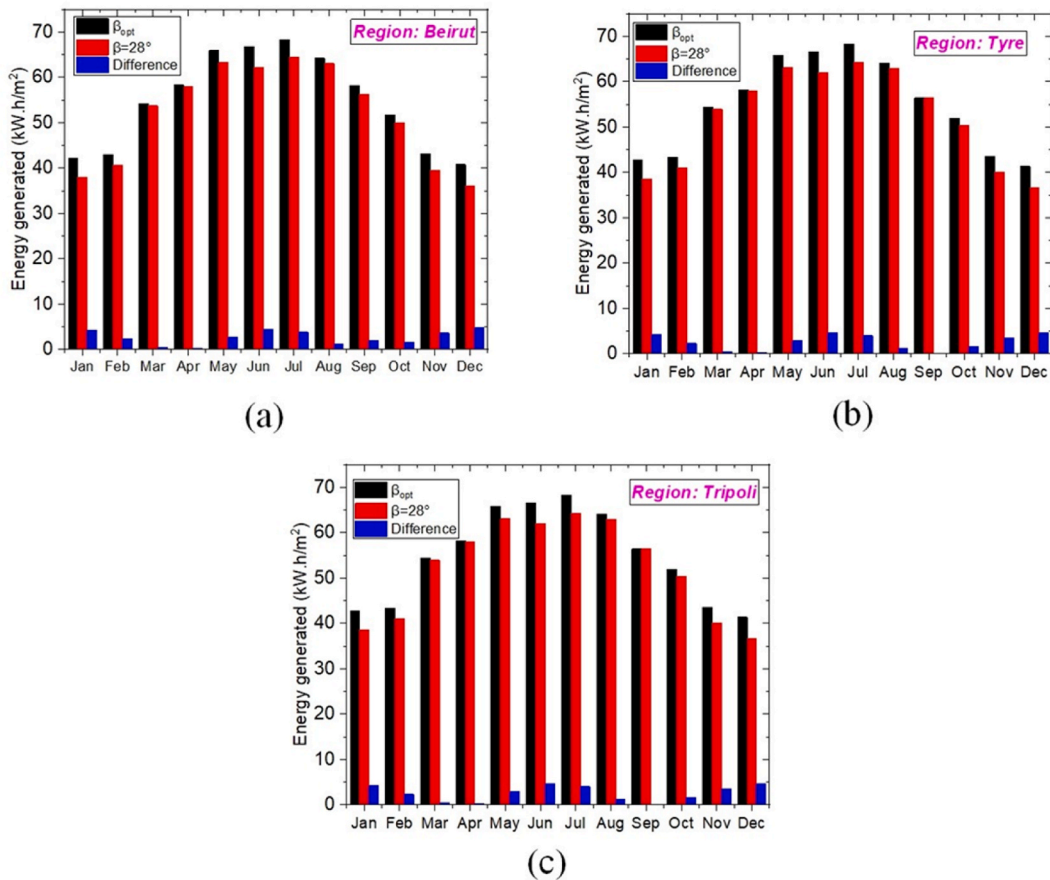


Fig. 6. Energy generated per square meter for the monthly variable optimum tilt angle and the fixed optimum angle over the year for (a) Beirut, (b) Tyre, and (c) Tripoli.

energy needed for water pumping during this period is 80 MW h. Additionally, as in the first case, assuming that CO₂ emissions from fossil fuel generation amount to 0.5 kg per kWh and that CO₂ emissions for 1 m² of solar panels are 10 kg per year, with a panel lifespan of 20 years, with the cost of 1 kWh from the national electricity grid was assumed to be \$0.20. Table 3 shows the cost required to generate 80 MW h using variable monthly, seasonal, and fixed yearly optimum tilt angles, compared to the cost of 80 MW h from the national electricity grid.

The results indicate that using a variable optimum tilt angle reduces the cost of 1 kWh by approximately 67 % compared to national grid electricity, and by about 1 % when comparing a monthly variable angle to a yearly fixed angle. In terms of emissions, using a solar power system reduces emissions by 89 % compared to electricity from the national grid, and by around 0.4 % when comparing a monthly variable tilt angle to a fixed tilt angle. If a 0.6 % annual deterioration in panel efficiency is considered, the cost of 1 kWh is reduced to approximately 63 % compared to national grid electricity, and emissions are reduced by around 84 %. Furthermore, if dust accumulation on the solar panels is taken into account—resulting in an estimated 10 % reduction in efficiency—the cost of 1 kWh drops to about 57 % of the national grid cost, and emissions are reduced by approximately 76 %.

5. conclusions

PV modules' tilt angles have an impact on the amount of sunlight that strikes their surface, directly impacting their electric efficiency generation. This study introduces a numerical systematic approach for determining the optimal tilt angle of PV panels in three Lebanese cities: Beirut, Tyre, and Tripoli. The method considers variable solar angles over different time periods – monthly, seasonal, and yearly – to maximize the energy output from solar panels. Additionally, this approach was constructed to be highly versatile and adaptable, enabling it to be implemented regardless of the geographical location, forming a valuable tool in enhancing the harvesting of solar energy and increasing the PV electric efficiency across different regions and at different operational periods.

Results have identified the optimum angle for both fixed and variable tilt angles (monthly or seasonally) and highlight the importance of using solar panels in reducing electricity costs and carbon emissions, contributing to a more cost-effective and sustainable solution. Additionally, the use of variable tilt angle – whether adjusted monthly or seasonally – enhances the solar power output of a solar panel by approximately 5 % compared to fixed PV tilt angle. Moreover, the utilization of solar panels reduces carbon

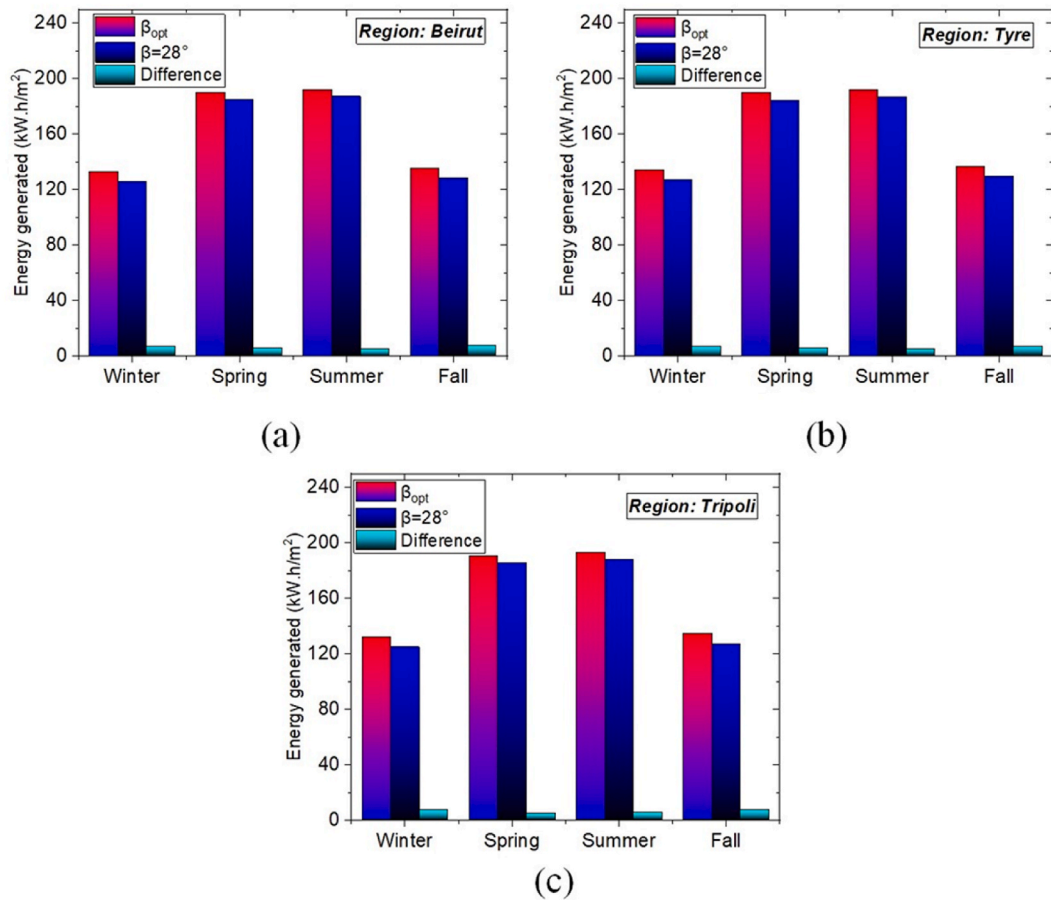


Fig. 7. Power generated per square meter for the seasonal optimum tilt angle and the constant yearly optimum angle in (a) Beirut, (b) Tyre, and (c) Tripoli.

Table 2

Cost required to generate 1 kWh with yearly fixed and variable monthly and seasonal optimum tilt angles in the three cities and compared to national electricity grid cost.

City	Beta (β)	Energy generated (kWh/year.m ²)	Needed PV Area to generate 5000 kWh (m ²)	System Cost per year (\$)	Cost of kWh (\$)	CO ₂ PV emissions (kg/year)	CO ₂ emission reduction (%)	Cost reduction (%)
Beirut	Fixed	625.33	8	99.95	0.02	79.96	96.8	90.01
	Varies	651.54	7.67	95.93	0.0192	76.74	96.93	90.41
	Seasonally							
	Varies	656.7	7.61	95.17	0.019	76.14	96.95	90.48
Tyre	Fixed	627.18	7.97	99.65	0.0199	79.72	96.81	90.03
	Varies	651.54	7.67	95.93	0.0192	76.74	96.93	90.41
	Seasonally							
	Varies	656.72	7.61	95.17	0.019	76.14	96.95	90.48
Tripoli	Fixed	623.79	8.02	100.19	0.02	80.16	96.79	89.98
	Varies	648.95	7.7	96.31	0.0193	77.05	96.92	90.37
	Seasonally							
	Varies	653.13	7.66	95.69	0.0191	76.55	96.94	90.43

Table 3

Cost required to generate 80 MWh with yearly fixed and variable monthly and seasonal optimum tilt angle for a water station in the three cities and compared to national electricity grid cost.

Water Station Location	Beta (β)	Energy generated (kWh/year.m ²)	Needed PV Area to generate 80 MWh (m ²)	System Cost per year (\$)	Cost of kWh (\$)	CO ₂ PV emissions (kg/year)	CO ₂ emission reduction (%)	Cost reduction (%)
Beirut	Fixed	187.05	427.69	5346.16	0.0668	4276.93	89.31	66.59
	Varies	191.85	416.99	5212.41	0.0652	4169.92	89.58	67.42
	Seasonally							
	Varies	193.49	413.46	5168.27	0.0646	4134.62	89.66	67.7
Tyre	Monthly							
	Fixed	186.61	428.7	5358.77	0.067	4287.02	89.28	66.51
	Varies	191.74	417.23	5215.4	0.0652	4172.32	89.57	67.4
	Seasonally							
	Varies	192.13	416.38	5204.74	0.0651	4163.79	89.59	67.47
Tripoli	Monthly							
	Fixed	187.41	426.87	5335.89	0.0667	4268.72	89.33	66.65
	Varies	192.93	414.66	5183.23	0.0648	4146.58	89.63	67.6
	Seasonally							
	Varies	194.53	411.25	5140.6	0.0643	4112.48	89.72	67.87
	Monthly							

emissions by about 80 % compared to national grid, and by 0.3 % when comparing fixed to variable tilt angles.

CRediT authorship contribution statement

Hassan Moussa Nahim: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tarek Ibrahim:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Jalal Faraj:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Samer Ali:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Mahmoud Khaled:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Mohamad Darwiche:** Writing – review & editing, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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